

CROSS-E-strong-L3-01 v1: L3 vs E-strong audit

ZFL Lab

September 3, 2026

INPUT: $C_{L^3} = \text{PROVEN}$: $T_* < \infty \implies \limsup_{t \rightarrow T_*} \|u(t)\|_3 = +\infty$ (ESS). $C_E^{weak} = \text{PROVEN}$:
 $T_* < \infty \implies \limsup_{t \rightarrow T_*} \int_0^t W = +\infty$. $C_E^{strong} = \text{OPEN}$: $T_* < \infty \implies \lim_{t \rightarrow T_*} \int_0^t W = +\infty$.
 $W := \int \omega^T S \omega$.

MISSION: Audit $C_{L^3} \Rightarrow C_E^{strong}$? or $C_E^{strong} \Rightarrow C_{L^3}$?

FORBIDDEN: $C_E^{weak} \implies C_E^{strong}$ without extra control (signed W obstacle). $C_{L^3} \implies C_E^{strong}$
 by heuristic growth. Any contradiction without proven bridge.

Analysis:

1. $C_{L^3} \Rightarrow C_E^{strong}$?

We have $|W| \leq C_S \|u\|_3 \|\nabla \omega\|_2^2$ upper bound only. No lower bound $W \gtrsim \|u\|_3$ available.
 $\limsup \|u\|_3 = \infty$ does not prevent $\int_0^t W$ oscillating, even with large negative parts, keeping $\lim$
 finite while $\limsup = \infty$. Hence: $C_{L^3} \Rightarrow C_E^{strong}$ NOT PROVEN.

2. $C_E^{strong} \Rightarrow C_{L^3}$?

Assume C_E^{strong} true: $T_* < \infty \implies \int_0^t W \rightarrow +\infty$. Enstrophy: $\frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 = W - \nu \|\nabla \omega\|_2^2$.
 If $\int W \rightarrow +\infty$, then $\|\omega\|_2$ blows up (up to dissipation), hence $T_* < \infty$. But $T_* < \infty \implies$
 $\limsup \|u\|_3 = \infty$ is ESS, independent theorem, already PROVEN. So C_E^{strong} does not prove C_{L^3} ;

they coexist under ESS. Logical implication: $C_E^{strong} \Rightarrow C_{L^3}$ NOT PROVEN as implication, but COMPATIBLE v

Signed obstacle: C_E^{weak} is $\limsup \int W = +\infty$. C_E^{strong} needs $\lim \int W = +\infty$. W changes sign,
 no control of negative part from L^3 alone. Thus gap remains.

OUTPUT:

C_{L^3} and C_E^{strong} COMPATIBLE / OPEN

$C_{L^3} \Rightarrow C_E^{strong}$ OPEN

$C_E^{strong} \Rightarrow C_{L^3}$ OPEN as proof, COMPATIBLE via ESS

No new C_F . State remains $C_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$, C_E^{strong} OPEN.

Next bridge would require lower bound on W or control of negative stretching.